A textbook-style derivation of the Lefschetz thimble for the simple harmonic oscillator

From the path integral to the rotated Gaussian contour

with mini examples of eigenvalues, eigenvectors, determinants, Jacobians, and matrix square roots

This note is designed to read like a gentle maths chapter rather than a code walkthrough. The aim is to show, carefully and concretely, why the harmonic oscillator path integral becomes easy on a Lefschetz thimble, and exactly where quantities such as eigenvalues, eigenvectors, and determinants enter.

What you will get from this PDF

- The continuum action and its discrete matrix form.
- The exact eigenvectors and eigenvalues of the discrete oscillator matrix.
- The one-mode steepest descent rotation, and why the angle is $\pi/4$.
- The full thimble in many variables.
- A fully worked numerical example with $N = 4$, $T = 1$, $m = \omega = \hbar = 1$.

1. The big idea

In a real-time path integral, each path contributes a complex phase $\mathrm{e}^{\mathrm{i}S/\hbar}$. These contributions behave like arrows in the complex plane. On the original real contour, neighbouring paths can point in very different directions, so the arrows cancel heavily. This is the sign problem.

A Lefschetz thimble is a deformed contour in complexified configuration space on which the exponent becomes maximally well behaved. On the correct thimble, the phase is constant and the magnitude decays like an ordinary Gaussian. For the simple harmonic oscillator, because the action is quadratic, the thimble can be written down exactly.

Key slogan

For a quadratic action, the thimble is found by diagonalising the quadratic form, then rotating each eigenmode so that the oscillatory factor $\mathrm{e}^{\mathrm{i}S/\hbar}$ turns into a decaying Gaussian.

2. The harmonic oscillator action

The continuum action for a particle of mass m in the potential $V(x) = \frac{1}{2}m\omega^2 x^2$ is

$$S[x] = \int_0^T \left(\frac{m}{2}\dot{x}(t)^2 - \frac{m\omega^2}{2}x(t)^2 \right) \mathrm{d}t.$$

The real-time propagator is formally

$$K(x_f, T; x_i, 0) = \int_{x(0)=x_i}^{x(T)=x_f} \mathcal{D}x \; \mathrm{e}^{\mathrm{i}S[x]/\hbar}.$$

To derive the thimble cleanly, it is easiest to discretise time. Divide $[0, T]$ into N equal pieces, each of width

$$a = \frac{T}{N}.$$

Write the path values as

$$x_0 = x_i, \quad x_1, \ldots, x_{N-1}, \quad x_N = x_f.$$

We will mostly focus on the case $x_i = x_f = 0$, because then the algebra is cleanest and the thimble geometry is easiest to see.

A standard discrete action is

$$S(\boldsymbol{x}) = \sum_{j=0}^{N-1} \frac{m}{2a}(x_{j+1} - x_j)^2 - \frac{am\omega^2}{2} \sum_{j=1}^{N-1} x_j^2,$$

where $\boldsymbol{x} = (x_1, \ldots, x_{N-1})^T$ contains the interior variables.

Strictly, real-time Gaussian integrals are defined with a tiny regulator $\omega^2 \to \omega^2 - \mathrm{i}\varepsilon$ or, equivalently, $S \to S + \frac{\mathrm{i}\varepsilon}{2}\int x^2 \mathrm{d}t$. That nudges the contour very slightly away from an exact 45° rotation. I will first derive the clean $\varepsilon \to 0^+$ formulas, then explain how the regulator modifies the angle.

3. Turning the action into a matrix

Because the action is quadratic, we can rewrite it as a matrix expression. Expand the kinetic part:

$$\sum_{j=0}^{N-1}(x_{j+1}-x_j)^2 = \sum_{j=0}^{N-1}\left(x_{j+1}^2 - 2x_{j+1}x_j + x_j^2\right).$$

If $x_0 = x_N = 0$, this simplifies to

$$\sum_{j=0}^{N-1}(x_{j+1}-x_j)^2 = 2\sum_{j=1}^{N-1}x_j^2 - 2\sum_{j=1}^{N-2}x_jx_{j+1}.$$

Hence

$$S(\boldsymbol{x}) = \left(\frac{m}{a} - \frac{am\omega^2}{2}\right)\sum_{j=1}^{N-1}x_j^2 - \frac{m}{a}\sum_{j=1}^{N-2}x_jx_{j+1}.$$

This is exactly

$$S(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T K\,\boldsymbol{x},$$

with the tridiagonal matrix

$$K = \begin{pmatrix} \frac{2m}{a} - am\omega^2 & -\frac{m}{a} & 0 & \cdots & 0 \\ -\frac{m}{a} & \frac{2m}{a} - am\omega^2 & -\frac{m}{a} & \ddots & \vdots \\ 0 & -\frac{m}{a} & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \frac{2m}{a} - am\omega^2 & -\frac{m}{a} \\ 0 & \cdots & 0 & -\frac{m}{a} & \frac{2m}{a} - am\omega^2 \end{pmatrix}.$$

So the path integral over the $N-1$ interior variables is a multidimensional oscillatory Gaussian.

> **Why this matters**
>
> Once you have $S(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T K\boldsymbol{x}$, the whole problem becomes a linear algebra problem. The thimble is controlled by the eigenvalues and eigenvectors of K.

4. Mini linear algebra toolkit

Before deriving the thimble, here are the bits of linear algebra that matter.

4.1 Eigenvalues and eigenvectors

If a matrix M satisfies

$$M\boldsymbol{v} = \lambda\boldsymbol{v},$$

then $\boldsymbol{v}$ is an eigenvector and λ is its eigenvalue. The matrix acts on that direction by simple scaling.

Tiny example. Let

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$M\begin{pmatrix}1\\1\end{pmatrix} = 3\begin{pmatrix}1\\1\end{pmatrix}, \qquad M\begin{pmatrix}1\\-1\end{pmatrix} = 1\begin{pmatrix}1\\-1\end{pmatrix}.$$

So the eigenvalues are 3 and 1. In the basis of these eigenvectors, the matrix is diagonal.

4.2 Determinants

For a 2×2 matrix,

$$\det\begin{pmatrix}a & b\\ c & d\end{pmatrix} = ad - bc.$$

The determinant measures how volumes scale under the linear map. For the matrix above,

$$\det M = 2 \cdot 2 - 1 \cdot 1 = 3.$$

Notice that

$$3 = 3 \times 1,$$

which is the product of the eigenvalues. This is true quite generally:

$$\det M = \prod_n \lambda_n.$$

4.3 Jacobians and change of variables

If $x = 2z$ in one variable, then $\mathrm{d}x = 2\,\mathrm{d}z$ and

$$\int f(x)\,\mathrm{d}x = \int f(2z)\,2\,\mathrm{d}z.$$

In many variables, if $\boldsymbol{x} = A\boldsymbol{z}$, then

$$\mathrm{d}^n x = (\det A)\,\mathrm{d}^n z.$$

If A is complex, then the phase of $\det A$ matters, not just its magnitude. This is exactly why a determinant appears when you parametrise a thimble by a matrix A.

4.4 Matrix square roots

If K is symmetric and positive definite, diagonalise it as

$$K = U\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)U^T,$$

with $U^T U = I$. Then

$$K^{-1/2} = U\,\mathrm{diag}\left(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_n}}\right)U^T.$$

This matrix will appear naturally in the thimble map.

5. The exact eigenmodes of the discrete oscillator

Because K is tridiagonal and translation-invariant in the bulk, its eigenvectors are discrete sine waves. Try

$$v_j^{(n)} = \sin\left(\frac{jn\pi}{N}\right), \qquad j = 1, \ldots, N-1.$$

Let us check this. For an interior index j,

$$(K\boldsymbol{v}^{(n)})_j = \left(\frac{2m}{a} - am\omega^2\right) v_j^{(n)} - \frac{m}{a} v_{j-1}^{(n)} - \frac{m}{a} v_{j+1}^{(n)}.$$

Use the identity

$$\sin((j-1)\theta) + \sin((j+1)\theta) = 2\sin(j\theta)\cos\theta,$$

with $\theta = n\pi/N$. Then

$$(K\boldsymbol{v}^{(n)})_j = \left[\frac{2m}{a} - am\omega^2 - \frac{2m}{a}\cos\left(\frac{n\pi}{N}\right)\right] v_j^{(n)}.$$

So the eigenvalue is

$$\lambda_n = \frac{2m}{a}\left(1 - \cos\frac{n\pi}{N}\right) - am\omega^2 = \frac{4m}{a}\sin^2\left(\frac{n\pi}{2N}\right) - am\omega^2, \qquad n = 1, \ldots, N-1.$$

After normalisation, these eigenvectors form an orthonormal matrix U, and therefore

$$K = U\,\mathrm{diag}(\lambda_1, \ldots, \lambda_{N-1})U^T.$$

If we define the mode coordinates by

$$\boldsymbol{x} = U\boldsymbol{q},$$

then the action factorises:

$$S = \frac{1}{2}\sum_{n=1}^{N-1} \lambda_n q_n^2.$$

This is the main simplification.

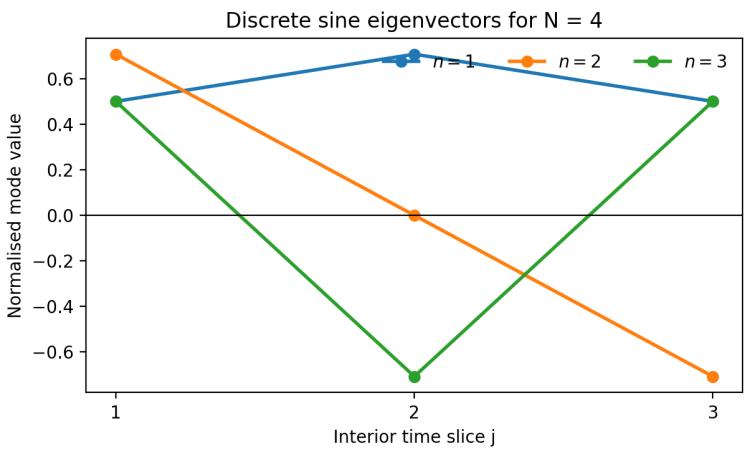

Figure 1: For $N = 4$, the eigenvectors are discrete sine waves. Diagonalising the action means rewriting the path in terms of these independent mode amplitudes.

> **A useful physical reading**
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> Each eigenmode is an independent quadratic degree of freedom. The many-dimensional thimble is just the product of the one-dimensional thimbles for these modes.

6. The one-mode thimble

Now take a single mode with action

$$S_n(q_n) = \frac{1}{2}\lambda_n q_n^2.$$

The corresponding factor in the path integral is

$$\exp\left(\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2\right).$$

Suppose first that $\lambda_n > 0$. If we keep q_n real, the exponent is purely imaginary and the integrand oscillates forever. To make it decay, rotate the contour by 45°:

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4} y_n, \qquad y_n \in \mathbb{R}.$$

Then

$$q_n^2 = \mathrm{e}^{\mathrm{i}\pi/2} y_n^2 = \mathrm{i} y_n^2,$$

and therefore

$$\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2 = \frac{\mathrm{i}}{2\hbar}\lambda_n(\mathrm{i} y_n^2) = -\frac{\lambda_n}{2\hbar} y_n^2.$$

So the oscillatory factor has become a decaying Gaussian:

$$\exp\left(\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2\right) \longrightarrow \exp\left(-\frac{\lambda_n}{2\hbar} y_n^2\right).$$

If instead $\lambda_n < 0$, write $\lambda_n = -|\lambda_n|$. Then the correct rotation is the opposite one,

$$q_n = \mathrm{e}^{-\mathrm{i}\pi/4} y_n,$$

which again gives

$$\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2 = -\frac{|\lambda_n|}{2\hbar} y_n^2.$$

Therefore the thimble direction for each mode is

$$q_n = \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\,\mathrm{sgn}(\lambda_n)} y_n, \qquad y_n \in \mathbb{R}.$$

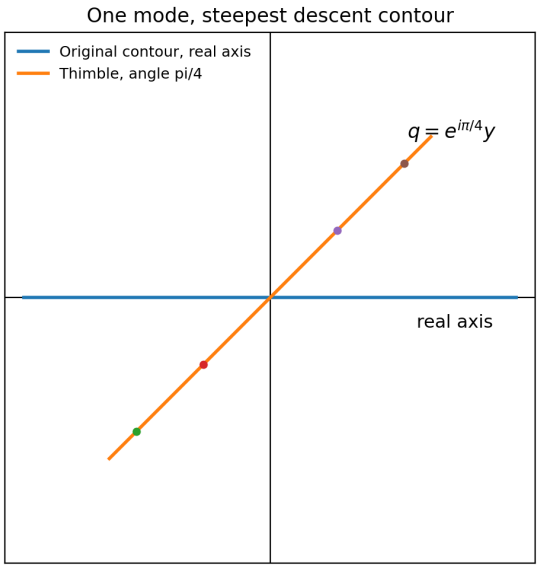

Figure 2: For one positive mode, the steepest descent contour is the line at angle $\pi/4$. Along that contour the exponent becomes real and negative.

A tiny numerical example

Take the toy integral

$$I = \int \mathrm{e}^{\mathrm{i}q^2}\,\mathrm{d}q.$$

Set $q = \mathrm{e}^{\mathrm{i}\pi/4}y$. Then $\mathrm{d}q = \mathrm{e}^{\mathrm{i}\pi/4}\mathrm{d}y$ and

$$I = \mathrm{e}^{\mathrm{i}\pi/4}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2}\,\mathrm{d}y = \mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\pi}.$$

This is the simplest example of a thimble calculation.

7. The full thimble for the oscillator

Put the mode rotations together. In mode space,

$$\boldsymbol{q} = D\boldsymbol{y}, \qquad D = \mathrm{diag}\Big(\mathrm{e}^{\mathrm{i}\frac{\pi}{4}\,\mathrm{sgn}(\lambda_1)}, \dots, \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\,\mathrm{sgn}(\lambda_{N-1})}\Big).$$

Returning to the original variables,

$$\boldsymbol{x} = U\boldsymbol{q} = UD\boldsymbol{y}, \qquad \boldsymbol{y} \in \mathbb{R}^{N-1}.$$

This is the geometric thimble. Along it,

$$\frac{\mathrm{i}}{\hbar}S(\boldsymbol{x}) = -\frac{1}{2\hbar}\sum_{n=1}^{N-1}|\lambda_n|y_n^2,$$

so the phase is constant and the magnitude decays.

A more convenient sampling parametrisation

It is often nicer to absorb the factors $|\lambda_n|$ into the coordinates. Define

$$y_n = \sqrt{\frac{\hbar}{|\lambda_n|}}\,z_n, \qquad z_n \in \mathbb{R}.$$

Then

$$\boldsymbol{x} = A\boldsymbol{z}, \qquad A = U\,\mathrm{diag}\left(\mathrm{e}^{\mathrm{i}\frac{\pi}{4}\,\mathrm{sgn}(\lambda_n)}\sqrt{\frac{\hbar}{|\lambda_n|}}\right).$$

With this choice,

$$\frac{\mathrm{i}}{\hbar}S(A\boldsymbol{z}) = -\frac{1}{2}\sum_{n=1}^{N-1}z_n^2.$$

So the exponent is exactly the standard Gaussian exponent.

If all eigenvalues are positive, this becomes especially neat:

$$A = \mathrm{e}^{\mathrm{i}\pi/4}\,U\,\mathrm{diag}\left(\frac{\sqrt{\hbar}}{\sqrt{\lambda_n}}\right)U^T = \mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\hbar}\,K^{-1/2}.$$

Then

$$A^T K A = \mathrm{i}\hbar\, I,$$

which is the matrix version of the one-mode statement.

Why the determinant appears

If you use $\boldsymbol{x} = A\boldsymbol{z}$ to parametrise the thimble, then

$$\mathrm{d}^{N-1}x = (\det A)\,\mathrm{d}^{N-1}z.$$

That factor is the Jacobian. Because A is complex, $\det A$ is complex too, and its phase is part of the final answer.

8. The regulator and the slightly shifted angle

With the usual regulator $\omega^2 \to \omega^2 - \mathrm{i}\varepsilon$, the eigenvalues become slightly complex:

$$\lambda_n \to \lambda_{n,\varepsilon} = |\lambda_{n,\varepsilon}|\mathrm{e}^{\mathrm{i}\phi_n}.$$

To make the exponent decaying, choose

$$q_n = \mathrm{e}^{\mathrm{i}\theta_n} y_n, \qquad \theta_n = \frac{\pi}{4} - \frac{\phi_n}{2}.$$

Then

$$\mathrm{i}\lambda_{n,\varepsilon}\mathrm{e}^{2\mathrm{i}\theta_n} = \mathrm{i}|\lambda_{n,\varepsilon}|\mathrm{e}^{\mathrm{i}\phi_n}\mathrm{e}^{\mathrm{i}\pi/2-\mathrm{i}\phi_n} = -|\lambda_{n,\varepsilon}|.$$

So the exact steepest descent contour is no longer at exactly 45°, but it is very close when ε is small.

This is why machine-learned thimbles for the regulated action often look like a nearly uniform 45° rotation, with a small correction in the precise scaling and phase of each mode.

9. Fully worked numerical example

Now let us do the whole thing explicitly for

$$N = 4, \qquad T = 1, \qquad m = 1, \qquad \omega = 1, \qquad \hbar = 1,$$

with endpoints $x_0 = x_4 = 0$. Then

$$a = \frac{T}{N} = \frac{1}{4}.$$

There are $N - 1 = 3$ interior variables: x_1, x_2, x_3.

9.1 The discrete matrix

The diagonal and off-diagonal entries are

$$\frac{2m}{a} - am\omega^2 = 8 - 0.25 = 7.75, \qquad -\frac{m}{a} = -4.$$

So

$$K = \begin{pmatrix} 7.75 & -4 & 0 \\ -4 & 7.75 & -4 \\ 0 & -4 & 7.75 \end{pmatrix}.$$

9.2 The eigenvalues

Using

$$\lambda_n = \frac{2m}{a}\left(1 - \cos\frac{n\pi}{N}\right) - am\omega^2, \qquad n = 1, 2, 3,$$

we get

$$\lambda_1 \approx 2.09314575, \qquad \lambda_2 = 7.75, \qquad \lambda_3 \approx 13.40685425.$$

All three are positive, so every mode rotates by $+\pi/4$.

9.3 The eigenvectors

The normalised discrete sine eigenvectors are

$$\boldsymbol{v}^{(1)} = \begin{pmatrix} 0.5 \\ 0.7071 \\ 0.5 \end{pmatrix}, \qquad \boldsymbol{v}^{(2)} = \begin{pmatrix} 0.7071 \\ 0 \\ -0.7071 \end{pmatrix}, \qquad \boldsymbol{v}^{(3)} = \begin{pmatrix} 0.5 \\ -0.7071 \\ 0.5 \end{pmatrix}.$$

Hence

$$U = \begin{pmatrix} 0.5 & 0.7071 & 0.5 \\ 0.7071 & 0 & -0.7071 \\ 0.5 & -0.7071 & 0.5 \end{pmatrix}.$$

9.4 The thimble map

Because all eigenvalues are positive and $\hbar = 1$,

$$A = \mathrm{e}^{\mathrm{i}\pi/4} K^{-1/2} = U \operatorname{diag}\left(\frac{\mathrm{e}^{\mathrm{i}\pi/4}}{\sqrt{\lambda_1}}, \frac{\mathrm{e}^{\mathrm{i}\pi/4}}{\sqrt{\lambda_2}}, \frac{\mathrm{e}^{\mathrm{i}\pi/4}}{\sqrt{\lambda_3}}\right) U^T.$$

Numerically,

$$A \approx \begin{pmatrix} 0.2975 + 0.2975\mathrm{i} & 0.1045 + 0.1045\mathrm{i} & 0.0435 + 0.0435\mathrm{i} \\ 0.1045 + 0.1045\mathrm{i} & 0.3409 + 0.3409\mathrm{i} & 0.1045 + 0.1045\mathrm{i} \\ 0.0435 + 0.0435\mathrm{i} & 0.1045 + 0.1045\mathrm{i} & 0.2975 + 0.2975\mathrm{i} \end{pmatrix}.$$

If $\boldsymbol{x} = A\boldsymbol{z}$ with $\boldsymbol{z} \in \mathbb{R}^3$, then

$$\frac{\mathrm{i}}{\hbar} S(A\boldsymbol{z}) = -\frac{1}{2}(z_1^2 + z_2^2 + z_3^2).$$

So on the thimble the path integral really has become a standard decaying Gaussian.

9.5 The determinant

The Jacobian is

$$\det A = \prod_{n=1}^{3} \frac{\mathrm{e}^{\mathrm{i}\pi/4}}{\sqrt{\lambda_n}} = \frac{\mathrm{e}^{3\mathrm{i}\pi/4}}{\sqrt{\lambda_1\lambda_2\lambda_3}}.$$

Now

$$\lambda_1\lambda_2\lambda_3 = \det K \approx 217.484375,$$

so

$$|\det A| = \frac{1}{\sqrt{217.484375}} \approx 0.06781.$$

The phase is $3\pi/4$. So

$$\det A \approx -0.04795 + 0.04795\mathrm{i}.$$

This number is not a decoration. It is the exact Jacobian factor that multiplies the Gaussian integral after the change of variables.

Mode n	λ_n	$1/\sqrt{\lambda_n}$	rotation angle
1	2.09314575	0.6911	$\pi/4$
2	7.75	0.3592	$\pi/4$
3	13.40685425	0.2731	$\pi/4$

10. What the derivation is really saying

There are three layers to keep distinct.

1. **Original variables.** The path is described by the interior values $\boldsymbol{x} \in \mathbb{R}^{N-1}$.
2. **Mode variables.** Diagonalising the quadratic form gives independent coordinates $\boldsymbol{q} = U^T\boldsymbol{x}$.
3. **Thimble variables.** Rotating and rescaling each mode gives real Gaussian variables $\boldsymbol{z}$ with $\boldsymbol{x} = A\boldsymbol{z}$.

The whole point of the thimble is that the difficult oscillatory weight

$$\mathrm{e}^{\mathrm{i}S(\boldsymbol{x})/\hbar}$$

becomes the easy Gaussian weight

$$\mathrm{e}^{-\frac{1}{2}\sum_n z_n^2},$$

after a clever complex change of variables.

One-line summary

For the quadratic harmonic oscillator, the exact Lefschetz thimble is obtained by diagonalising the discrete Hessian K, rotating each eigenmode by the steepest descent angle, and optionally rescaling by $1/\sqrt{|\lambda_n|}$ so the exponent becomes a standard Gaussian.

11. Common questions and confusions

Why is the rotation exactly $\pi/4$? Because multiplying by $\mathrm{e}^{\mathrm{i}\pi/4}$ squares to i, and the extra factor of i turns the oscillatory exponent into a negative real exponent.

Why do eigenvectors matter at all? Because the original coordinates are coupled. The eigenvectors are the special combinations of time slices that decouple the action into independent quadratic modes.

What is the determinant doing? It is the Jacobian of the change of variables. Without it, the transformed integral would not equal the original one.

What changes when the endpoints are not zero? Then the action has linear terms as well. One first shifts by the classical path, and the quadratic fluctuation matrix is still the same tridiagonal matrix. So the thimble for fluctuations is unchanged.

What if an eigenvalue is negative? Then that mode rotates by $-\pi/4$ rather than $+\pi/4$. In longer-time problems this is related to caustics and Maslov phases.

Is the thimble the same thing as a Gaussian proposal distribution? Not quite. The thimble is the geometric contour in complexified space. A Gaussian proposal distribution is one convenient way of parametrising and sampling points on that contour.

12. Final cheat sheet

Continuum action	$S[x] = \int_0^T \left(\frac{m}{2}\dot{x}^2 - \frac{m\omega^2}{2}x^2\right)\mathrm{d}t$
Discrete action	$S(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T K \boldsymbol{x}$
Matrix entries	$K_{jj} = \frac{2m}{a} - am\omega^2$, $K_{j,j+1} = K_{j+1,j} = -\frac{m}{a}$
Eigenvectors	$v_j^{(n)} \propto \sin(jn\pi/N)$
Eigenvalues	$\lambda_n = \frac{2m}{a}(1 - \cos(n\pi/N)) - am\omega^2$
One-mode thimble	$q_n = \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\operatorname{sgn}(\lambda_n)} y_n$
Scaled map	$x = Az$, $A = U \operatorname{diag}(\mathrm{e}^{\mathrm{i}\frac{\pi}{4}\operatorname{sgn}(\lambda_n)}\sqrt{\hbar/\lvert\lambda_n\rvert})$
Gaussian form on thimble	$\mathrm{i}S(Az)/\hbar = -\frac{1}{2}\sum_n z_n^2$
Jacobian	$\mathrm{d}^{N-1}x = (\det A)\,\mathrm{d}^{N-1}z$
Positive definite case	$A = \mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\hbar}\,K^{-1/2}$

The natural next companion note is a second PDF that starts from this derivation and then explains how the determinant, Gaussian sampling, importance weights, and effective sample size fit together for Monte Carlo on the thimble.